MEHRAN UNIVERSITY OF ENGINEERING AND TECHNOLOGY, JAMSHORO

DEPARTMENT OF ELECTRONIC ENGINEERING
INTEGRATED ELECTRONICS # 10

Roll No:	**Date:**
Checked by:	**Score:**

LAB PERFORMANCE INDICATOR	SUBJECT KNOWLEDGE	DATA ANALYSIS AND INTERPRETATION	ABILITY TO CONDUCT EXPERIMENT	PRESENTATION	CALCULATION AND CODING	OBSERVATION/ RESULTS	SCORE

OBJECT: To examine the input/output characteristics, voltage transfer characteristics, fan-out, propagation delay time of a CMOS gate

REQUIRED COMPONENTS:

i. Breadboard
ii. Few connecting hard wires.
iii. Few connecting LEDs.
iv. 4069 CMOS inverter

REQUIRED EQUIPMENTS:

i. Oscilloscope
ii. Dc power supply

CMOS CHARACTERISTICS

CMOS logic is exemplified by its extremely low power consumption and high noise immunity. Hence, it is prevalently used in devices demanding low power dissipation, such as digital wristwatches and other battery powered devices, or in devices operated in noisy environments, such as industrial plants.

The **voltage transfer curve** for a typical CMOS logic gate is shown in Figure 3.1. Note that the curves in the transition region are almost vertical. This narrow transition region is the reason for CMOS logic's high noise immunity. Not much voltage range is covered in the transition from one state to the other. In contrast to TTL devices, the threshold voltage depends on the supply voltage and is approximately half the supply voltage.

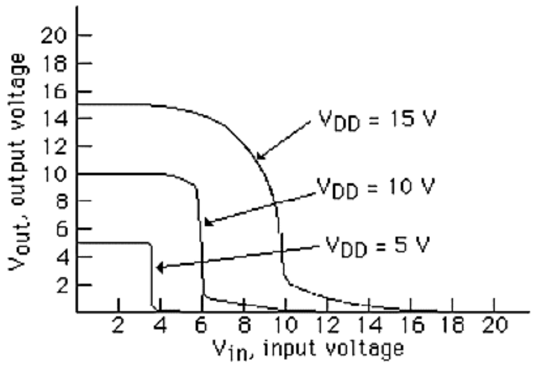


Figure 3.1: CMOS Voltage Transfer Curve

As with TTL logic, current spiking occurs during switching. Hence, bypass capacitors are used in CMOS logic design as well. However, they are not as critical as in TTL logic design because of CMOS's high noise immunity.

Whereas the typical quiescent (static) power dissipation (power dissipation of a device that is not changing logic states) of TTL IC's was about 40 mW, the power dissipation of CMOS IC's are typically 25 nW. However, as the frequency of switching increases, dynamic power dissipation becomes important, as illustrated in Figure 3.2. Above 1 MHz, the dynamic power dissipation predominates and can exceed TTL power dissipation

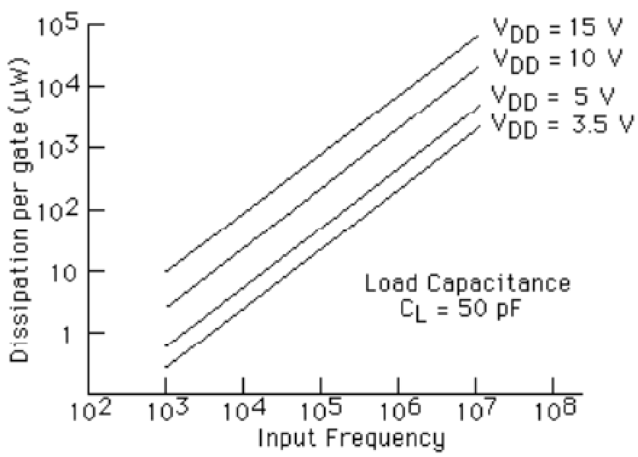


Figure 3.2: Dynamic Power Dissipation

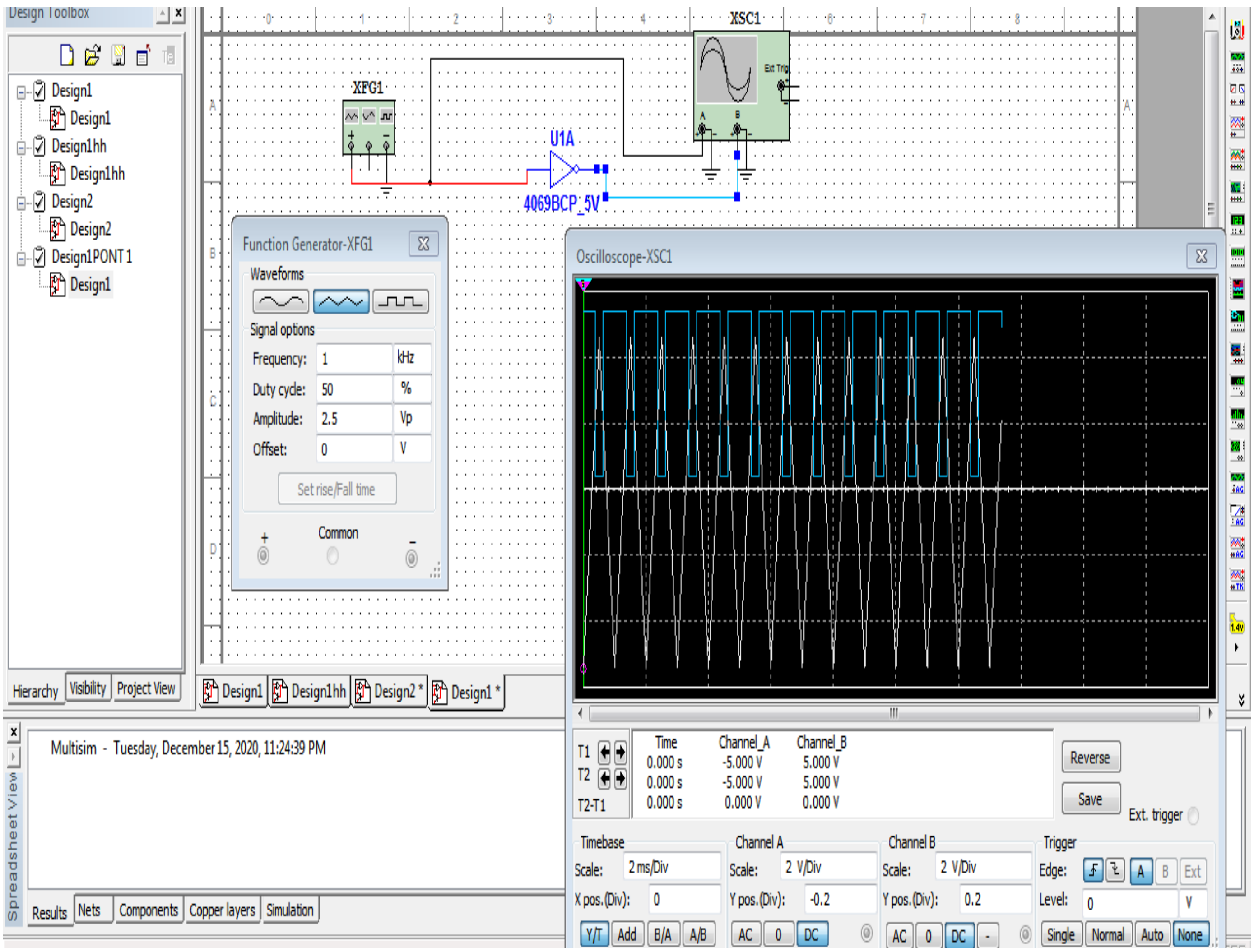
The **propagation delay** times for CMOS devices are relatively long due to their high output impedance. Typical delay times are 60 nsec for VDD = 5 V, and 25 nsec for VDD = 10 V. Doubling the supply voltage more than doubles the speed of a CMOS gate. The rise and fall transition times are typically 70 nsec. For VDD = 5 V. Thus CMOS devices operate significantly slower than TTL devices.

The HIGH and LOW **noise margins** for VDD = 5 V, are VIL - VOL = 1.5 - 0.01 ≈ 1.5 V, and VOH - VIH = 4.99 V - 3.5 V ≈ 1.5 V. In general, noise immunity is a minimum of 30% of VDD, and typically 45% of VDD. Thus, CMOS devices are good in noisy environments such as automobiles and industrial plants. The HIGH and LOW noise margins are essentially equal because the output impedance, output voltage, and threshold voltages are symmetrical with respect to the supply voltage.

The **fan-out** of CMOS devices is usually greater than 50 because the input current requirement of CMOS logic is nil (~pA). However, current is required to charge and discharge the capacitance of CMOS inputs during logic transitions. Hence, the greater the fan-out the longer the propagation delay. For example, with VDD = 5 V, the propagation delay will increase from 60 nsec when the output drives 1 input to 150 nsec when the output drives 10 inputs. As a rule of thumb, you can assume the load will be 5 pF per CMOS input plus 5 to 15 pF for stray wiring capacitance.

PROCEDURE

1. Use the oscilloscope (in the X-Y mode) to plot the transfer characteristics of the 4069 CMOS inverter. Use a triangularwave input of 5 Vp-p at 1 kHz (TTL compatible).



2. Label the regions of operation (cutoff, active, saturation) on the transfer characteristics. Also determine V_{IL}, V_{IH}, V_{OL} and V_{OH} from the curve.
3. Connect the circuit as shown in figure 3.3. Measure the voltage at the input of the inverter (V_{IH}). Using your measured voltage and Ohm's law, calculate the current in the resistor. This current is the input high current I_{IH}.

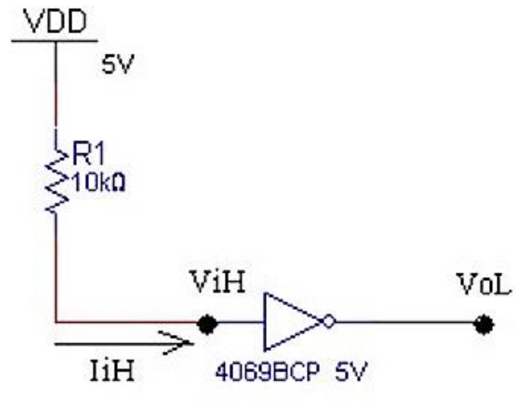
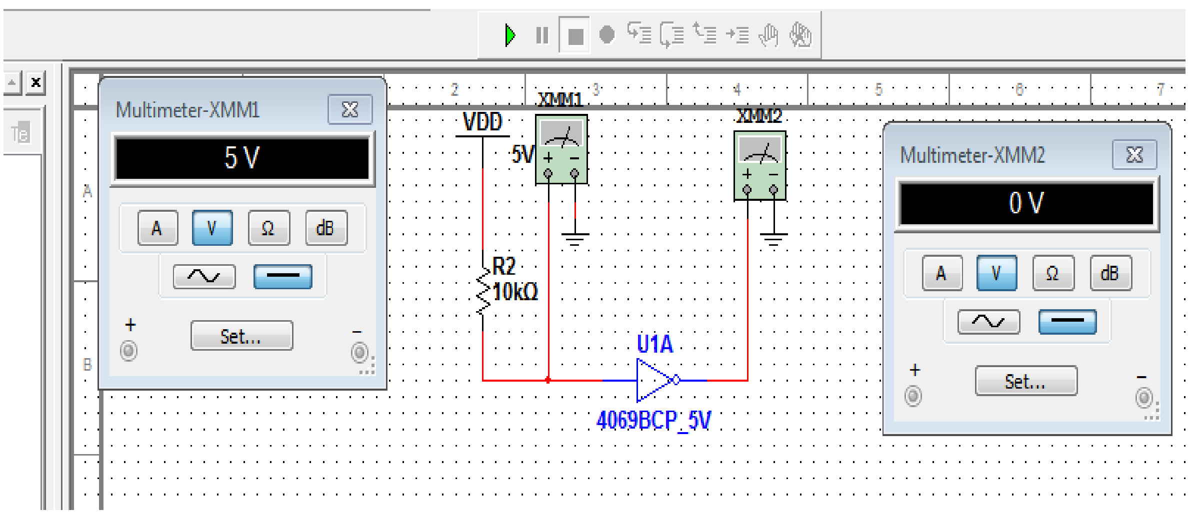


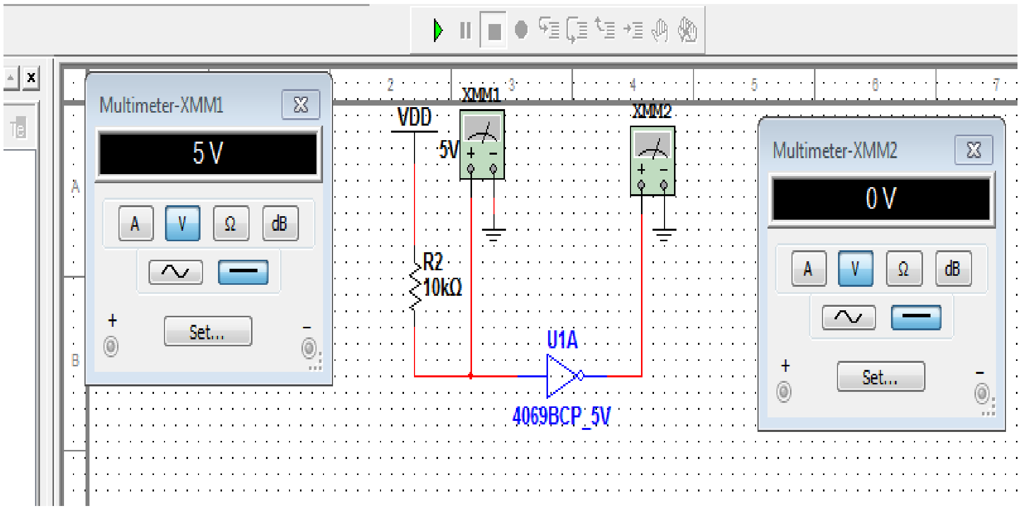
Figure 3.3



According ohm's law: V=IR => I=V/R

I=5/10k = 0.5mA

4. Measure the output voltage of the gate. Since the input is high, the output will be low. This voltage is called V_{OL}.



5. Connect the circuit as shown in figure 3.4. Measure V_{IL}. and V_{OH}. Calculate I_{IL} and I_{OH} applying Ohm's law.

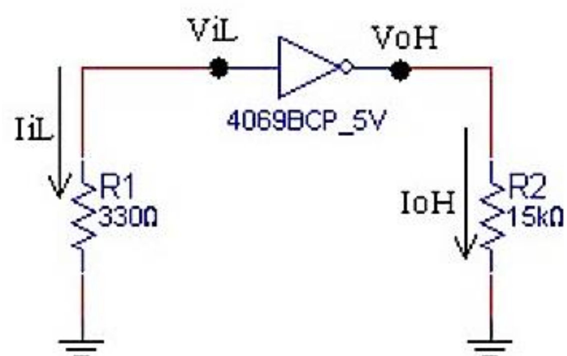
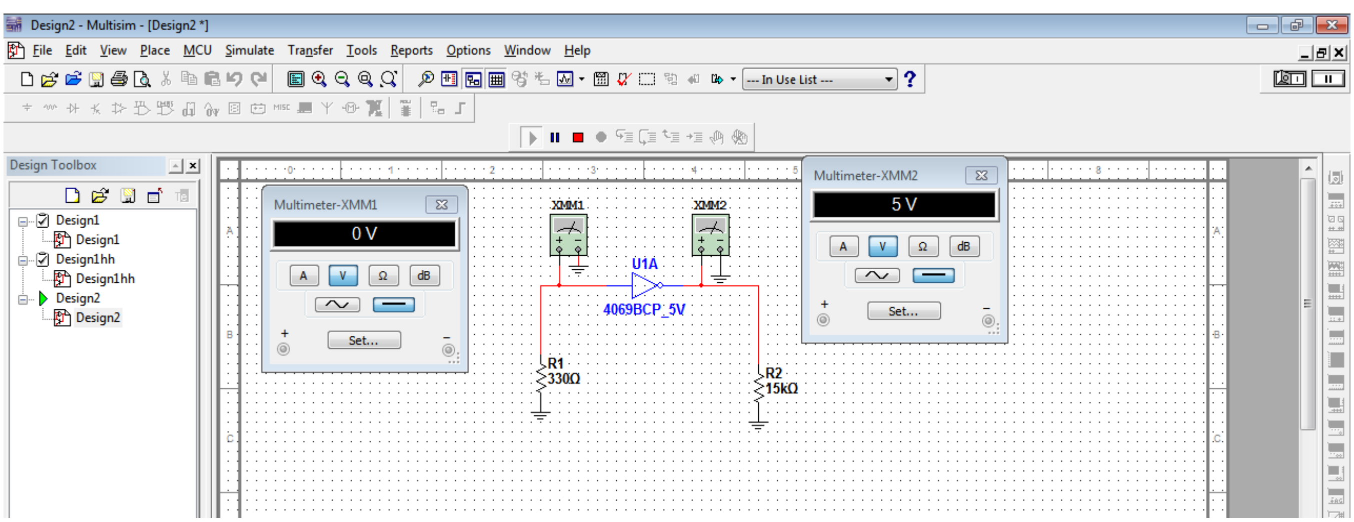


Figure 3.4



$I_{IL} = 0/330 = 0mA$; $I_{OH} = 5/15k = 0.33mA$

6. Compare your measured values with values in data sheet and calculate the low and high level noise margins.
7. The ring oscillator circuit shown in figure 3.5 is often used to measure the average propagation delay time of a logic gate. Draw 1st and last inverter waveforms and find out propagation delay.

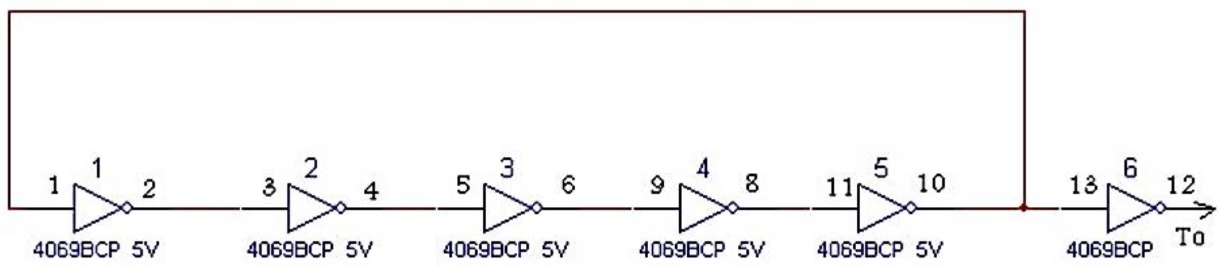
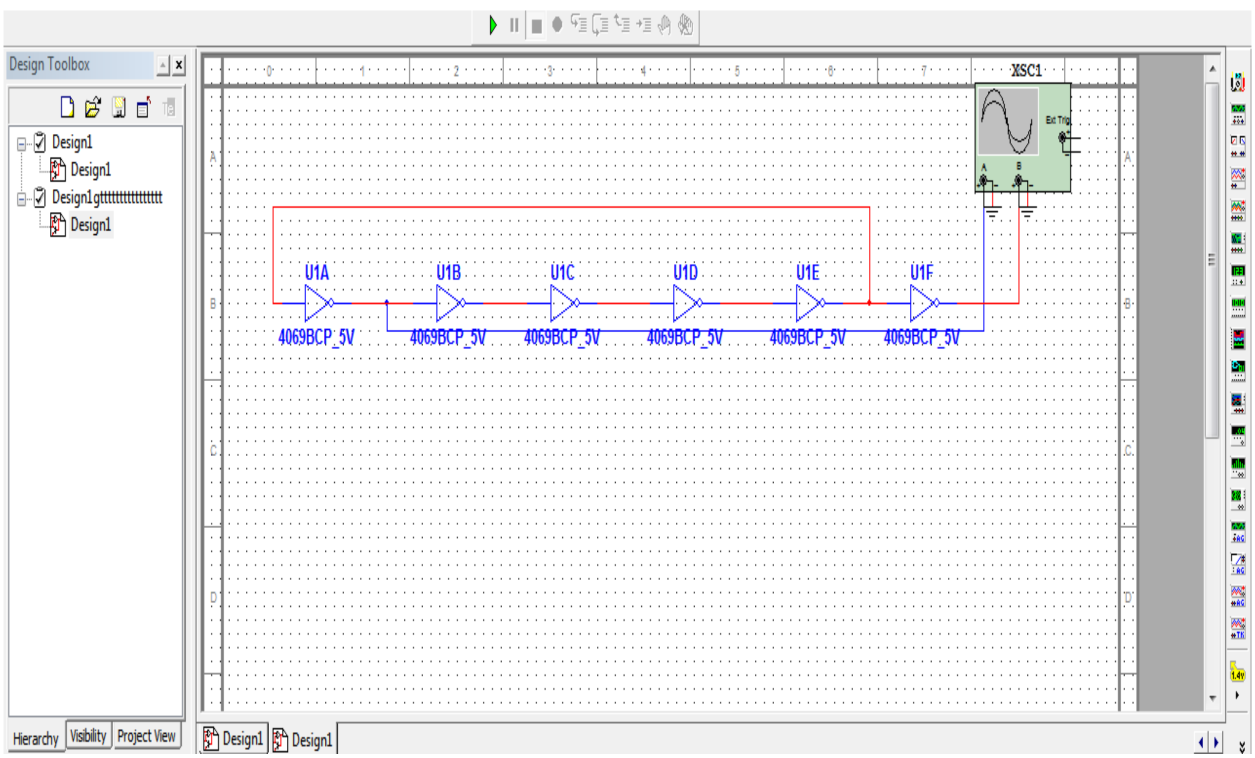


Figure 3.5



8. Connect the circuit as shown in figure 3.6. Measure the output voltage V_O when the fan-out is 1, 2, 3, 4 and 5 gates. Using these measurement determine the maximum number of gates which can be connected at the output of the driver gate such that $V_O \geq V_{IH}$ and $V_O \leq V_{IL}$.

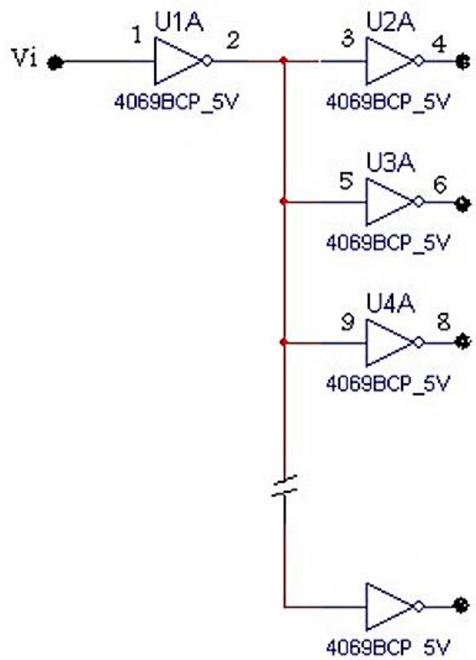
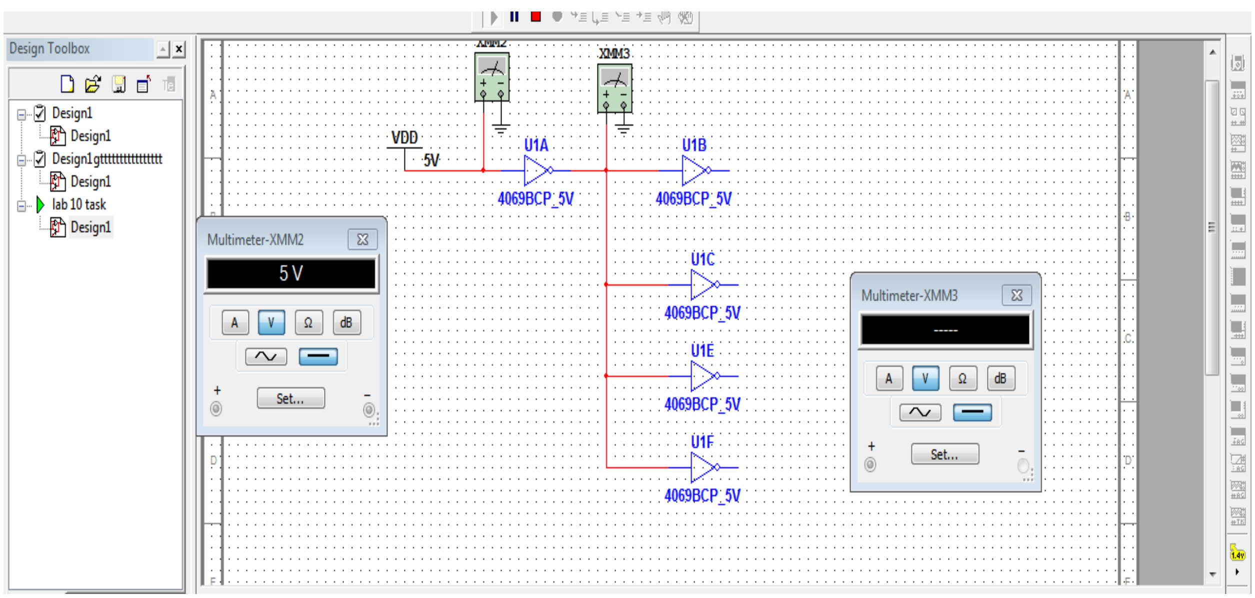


Figure 3.6



9. Make a complete analysis report, in which you state clearly a comparison between CMOS Inverter and TTL inverter characteristics.

Specifications	TTL	CMOS
Components	Passive Elements & Transistors	MOSFETs
Fan-out	10	>50
Noise Immunity	Strong	Extremely Strong
Noise Margin	Moderate	High
TPD in ns	1.5 to 30	1 to 210
Power/Gate in mWatt	10	0.0025

THE END